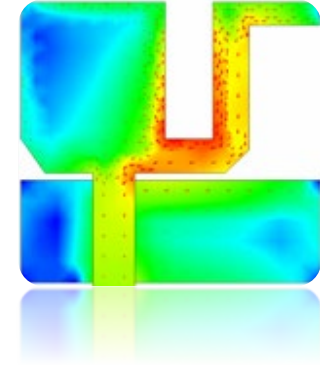




Antenna Engineering

Zhi Ning Chen

National University of Singapore

Assignment I

1. All solutions must be in A4 paper in PDF
2. Submission to folder for Assignment I thru Canvas (only once)
3. Submission deadline is 29 September 2024
4. File name: AI-(Student ID)-(Student Name)

Assignment I-Q1

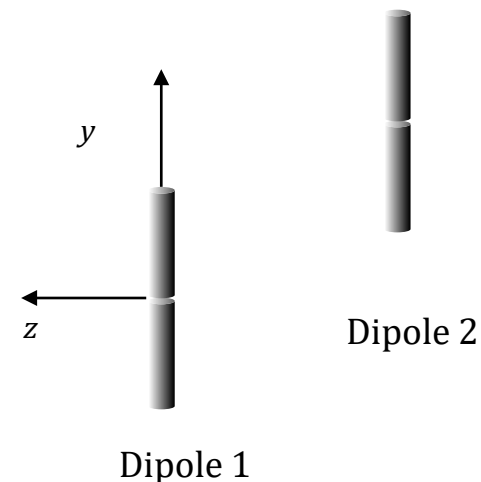
Q1: The uniform line electric current I flowing z -axis over an electrically thin ideal dipole is centered on the origin of a Cartesian coordinate system. Its length is $L=0.01 \lambda$.

- a) Find electric and magnetic fields in far-field zone;
- b) Sketch the radiation patterns in x - z and x - y planes;
- c) Find complex power density and radiated power;
- d) Find radiation resistance;
- e) Find directivity and gain if the ohmic loss of the radiator is $1(\Omega)$;
- f) Find the radiation efficiency, antenna efficiency and realized gain if the impedance matching between the antenna and a feed is $|S_{11}|=-10(\text{dB})$; and
- g) Find the distance of a far-field zone.

Assignment I-Q2

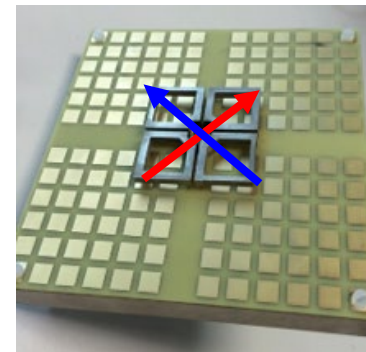
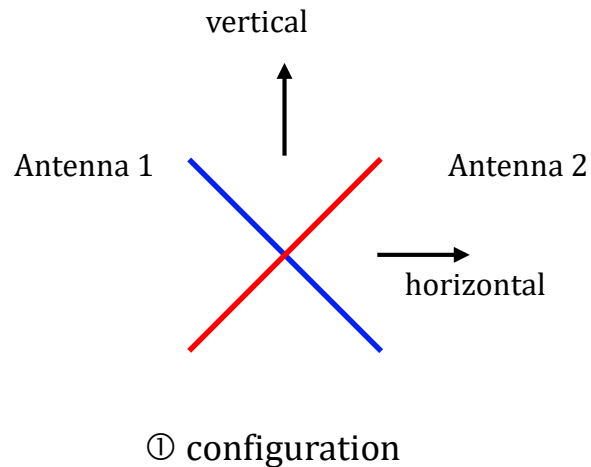
Q2. Consider two identical center-fed short dipoles operating at $f(\lambda)$, fed by two identical sources with the same voltage and phase. Dipole generates the electric field $\mathbf{E}$ at a distance r along z -axis in their far-field zone when the center of at least one dipole is located at the origin of coordinate system and along y -direction as shown in figure.

- To design an antenna system with the two dipoles which generating RHCP wave along z -direction at the distance r without use of any additional circuits
- Find the total electric field at the point of $z=r$.
- Find the total electric field at the point of $-z$, $|-z|=r$.
- Determine the polarization in $-z$ direction.



Assignment I-Q3

Q3. Explain the reason that the antennas used in mobile communication base-stations are installed at the same position with orthogonal polarization, and two of them are oriented in $\pm 45^\circ$ directions rather than horizontal and vertical directions as shown in figures



② prototype

$\pm 45^\circ$ cellular base-station antennas